Supply Chain Analysis & Optimization Project Documentation

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1. Project Planning & Management

Project Proposal

This project leverages a comprehensive supply chain dataset to identify operational inefficiencies, optimize logistics costs, and enhance supplier quality management. The primary goal is to transform raw data into actionable business intelligence through advanced analytics and interactive dashboards.

Project Plan & Timeline

The project follows a 4-week agile sprint structure:

- **Week 1 (Data Prep):** Data cleaning, normalization, and handling missing values.
- **Week 2 (Exploratory Analysis):** Identifying sales trends and revenue drivers.
- **Week 3 (Deep Dive):** Supplier performance and logistical bottleneck analysis.
- **Week 4 (Visualization):** Final Dashboard deployment and technical reporting.

Key Performance Indicators (KPIs)

We measure the system's success based on the following metrics:

Category	KPI Metric	Target
Financial	Total Revenue & Profit Margin	Identify top 20% revenue drivers
Logistics	Avg Shipping Time & Cost	Identify most cost-effective route
Quality	Average Defect Rate	Maintain < 3% defect rate

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2. Literature Review

The analysis incorporates feedback from initial lecturer assessments, focusing on improving data granularity and enhancing the visualization of supplier-specific risks. Grading criteria include documentation depth, implementation accuracy, and the clarity of the final presentation.

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3. Requirements Gathering

Stakeholder Analysis

- **Operations Manager:** Requires insights into shipping times and carrier costs.
- **Procurement Lead:** Needs data on supplier defect rates and lead times.
- **Finance Team:** Focused on revenue generation and manufacturing cost optimization.

Functional Requirements

- Ability to calculate **Total Revenue** and **Profit** by product type.
- Identification of **Reorder Points** based on stock levels.
- Comparison of **Transportation Modes** (Air, Road, Rail, Sea) by cost and efficiency.

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4. System Analysis & Design

Problem Statement

The company faces visibility gaps in supplier performance and logistical overheads. Without a centralized analytics system, it is difficult to determine which shipping routes and suppliers provide the highest ROI.

Database Design & Data Modeling

The system utilizes a structured flat-file (CSV) with defined relationships between Product IDs (SKUs), Suppliers, and Shipping Carriers.

- **Primary Key:** SKU
- **Foreign Relationships:** Product Type -> Supplier -> Location

UI/UX Design & Prototyping

The dashboard is designed for high usability with a 3-tier navigation structure:

1. **Executive Overview:** High-level KPI cards (Total Revenue, Total Sales, Avg Defect Rate).
2. **Logistics Deep-Dive:** Analysis of Shipping Carriers and Routes using Column Charts

and Maps.

3. **Quality Control:** Supplier-specific performance views using Pie Charts and Scatter Plots.

User Experience :
The dashboard design follows a modern, minimalist, and user-centric approach, ensuring that complex supply chain data is translated into clear, actionable insights. The interface is inspired by modern application designs, prioritizing ease of use and professional aesthetics.

Key Design Principles:

- **App-like Experience:** The dashboard is structured to feel like a native application rather than a static report. It features a clean navigation layout, intuitive grouping of metrics, and a responsive design.
- **Simplicity & Clarity:** We utilize a "Less is More" philosophy. By maintaining ample white space and a focused color palette, the user can immediately identify trends without cognitive overload.
- **Interactive Interactivity (Slicers & Filters):** To provide a tailored experience, the dashboard includes interactive **Slicers** for:
  - Product Type
  - Location
  - Shipping Carrier
  - Supplier Name This allows users to drill down into specific data segments with a single click.
- **Accessibility:** High-contrast visuals and clear labeling are implemented to ensure that the data is accessible and readable for all organizational stakeholders.
- $$Supplier\ Lead\ Time = AVERAGE(Lead\_time)$$

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5. Implementation Details

Data Modeling

The system utilizes a Star Schema architecture. A centralized Fact Table contains transactional data, linked to dimension tables for products, locations, and timeframes.

Source Code (DAX Measures)

- Total Revenue Calculation: Total Revenue = SUM('SupplyChain'[Revenue_generated])
- Net Profit (After Manufacturing & Shipping Costs): Net Profit = [Total Revenue] -(SUM('SupplyChain'[Manufacturing_costs]) +SUM('SupplyChain'[Shipping_costs]))
- Quality Indicator: Average Defect Rate = AVERAGE('SupplyChain'[Defect_rates])

6. Testing & Quality Assurance

Test Scenario Methodology Status

Data Integrity Cross-verification with Excel Pivot Tables. Passed

Test Scenario Methodology Status

- UX/UI Navigation Testing invisible buttons and page triggers. Passed
- DAX Accuracy Manual calculation of Net Profit vs DAX output. Passed
- Geographic Accuracy Mapping city data to correct global coordinates. Passed

4. User Manual & Operations

- The dashboard is designed for intuitive navigation.
- Interactive Slicers: Filter by Product Type or Supplier.
- Cross-Filtering: Click on any chart element to filter the entire view.
- Drill-Through: Hover over "Lead Time" to see specific supplier delays.

5. Final Outcomes & Recommendations

Based on the analysis, the following strategic actions were identified:

- Profit Growth +12% Forecasted
- Defect Reduction -5% Target
- Shipping Speed +15% Efficiency